

The Junior exam consists of problems **1 - 30**. The Senior exam consists of problems **11 - 40**. Students had **75 minutes** to complete the exam.

Problem 1.

There are nine integers on a blackboard whose average is 10. Ruhi chooses three integers and adds 5 to each of them. Then she subtracts 7 from all the other numbers. What will their new average be?

- A 1
- B 3
- C 5
- D 7
- E 9

Answer: D

Problem 2.

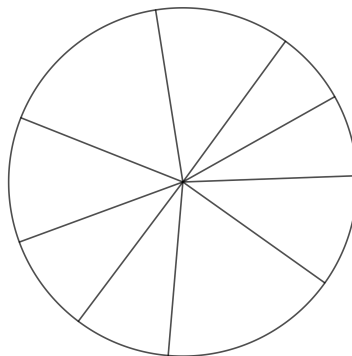
In how many ways can you choose four of the numbers 2, 3, 5, 7, 11 such that the product of two of them differs by 1 from the product of the other two?

- A 1
- B 2
- C 3
- D 4
- E 5

Answer: B

Problem 3.

A pizza is divided into 9 slices. How many possibilities are there to add pineapple to 4 of the slices in such a manner that no two pineapple slices are adjacent?



- A 1
- B 3
- C 5
- D 7
- E 9

Answer: E

Problem 4.

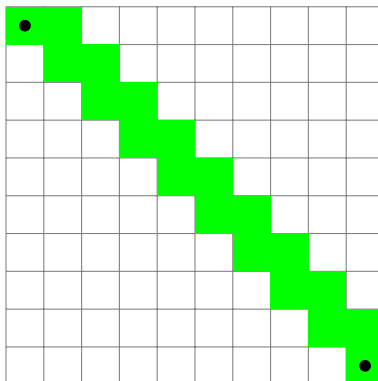
What is the product of the largest and second smallest number in the following list: $1 - \frac{1}{3}, 1 - \frac{1}{4}, 1 - \frac{1}{2}, 1 - \frac{1}{7}, 1 - \frac{1}{6}$?

- A $\frac{5}{12}$
- B $\frac{3}{7}$
- C $\frac{5}{9}$
- D $\frac{4}{7}$
- E $\frac{9}{14}$

Answer: D

Problem 5.

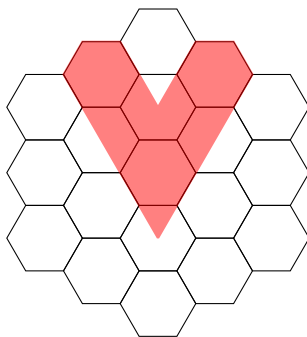
Turbo the snail is sitting on the upper left corner of the following 10×10 grid and she wants to go to the lower right corner by moving on adjacent tiles, to the right and downwards only. Jerry can place monsters on some tiles, which then Turbo is not allowed to visit. How many monsters does Jerry need to place at least to force Turbo to follow the green path?



Answer: 17

Problem 6.

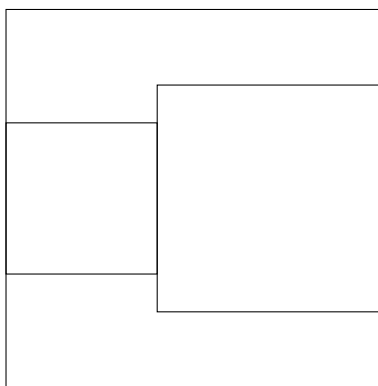
Each hexagon of this beehive has area 6. What is the area of the heart-shaped region?



Answer: 30

Problem 7.

Given that the two smaller squares have perimeters 2 and 3, what is the perimeter of the largest square in the image below?



Answer: 5

Problem 8.

What is the smallest positive integer n such that an $n \times n$ grid can be tiled with L-triominoes without overlap?

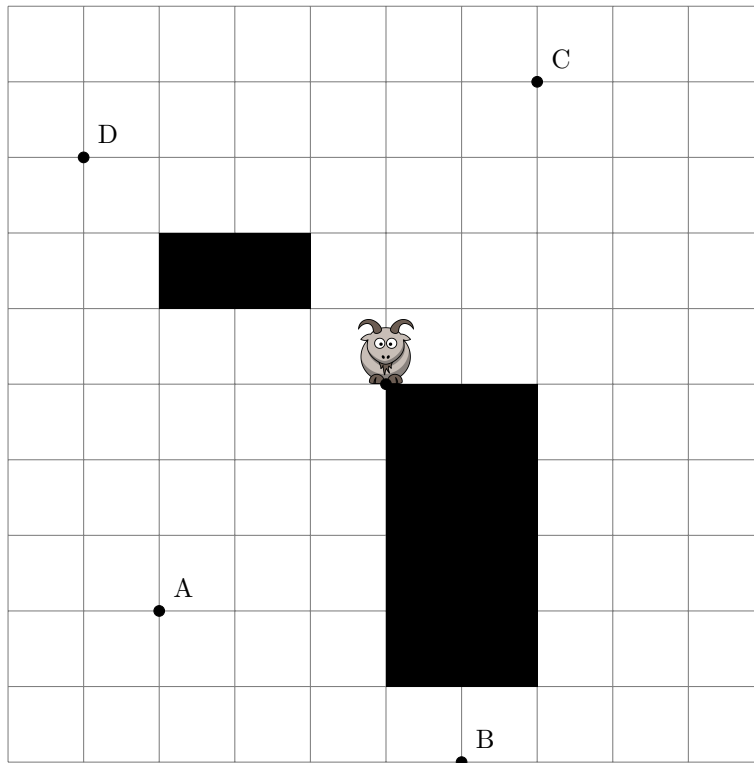


Answer: 6

Problem 9.

Koza the goat is attached to a pole in the centre of the 10×10 grid by a rope of length 5. Within that range it can move freely but neither Koza nor the rope can pass through the buildings in black. Which points can it reach?

- A
- B
- C
- D



Answer: A, C

Problem 10.

The Cantonal Bank of St. Gallen issues coins of values 1 franc, 11 francs, 101 francs and 1001 francs. Felix withdraws 2024 francs of pocket money from the bank. Which of the following statements are necessarily true?

- A Felix is given four kinds of coins.
- B Felix is given least 4 coins.

C Felix is given an odd number of coins.

D The Cantonal Bank of St. Gallen has multiple ways to pay Felix.

Answer: B, D

Problem 11.

A number n is written on a blackboard. In a move, Emily does the following: if n is even, she replaces it with $2n + 3$. If n is odd, she replaces it with $\frac{n+1}{2}$. If the initial number is 2024, what number is written on the blackboard after 10 moves?

A 2024

B 2034

C 2044

D 4071

E 4075

Answer: B

Problem 12.

Notice how $59 = 5 \cdot 9 + 5 + 9$. How many other two-digit numbers are equal to the product of their digits plus the sum of their digits?

A 8

B 9

C 10

D 11

E 12

Answer: A

Problem 13.

Among Anna, Bea, Clemens, Dimitri and Elia, two always lie and three always tell the truth. Here is their conversation:

- Anna: Bea is a liar.
- Bea: Elia and Clemens are liars.
- Clemens: Dimitri is a liar.
- Dimitri: Among Anna and Clemens there is exactly one liar.
- Elia: Anna and Dimitri either both tell the truth or both lie.

Who are the two liars?

- A Anna and Clemens.
- B Anna and Elia.
- C Bea and Clemens.
- D Bea and Dimitri.
- E Dimitri and Elia.

Answer: C

Problem 14.

Lena wants to cut a rectangular cake into rectangular pieces, so she only cuts horizontally or vertically, from one side of the cake all the way to the opposite side. At least how many cuts are needed in order to get exactly 180 pieces?

- A 24
- B 25
- C 26
- D 27
- E 28

Answer: B

Problem 15.

Mathys wants to make a five-digit number n with the digits 2, 3, 4, 6, 8 such that:

- n is even.
- The second digit of n is the product of the first and the third digit.
- The fourth digit of n is the product of the third and the fifth digit.

What is n ?

Remark: Digits are counted from the left.

Answer: 36284

Problem 16.

Which is the largest positive integer whose digits both add up and multiply to the number 8?

Answer: 22211

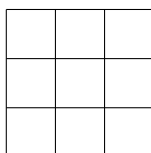
Problem 17.

Filling in the numbers 1, 2, 3, 4 into the boxes using each exactly once, what is the maximal possible value of $\square \cdot (\square + 2 \cdot \square) + \square$?

Answer: 33

Problem 18.

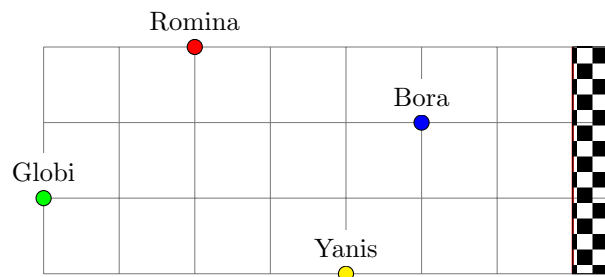
In how many ways can we colour the squares of a 3×3 board black and white such that any 2×2 region of the board consists of exactly 1 black and 3 white squares?



Answer: 8

Problem 19.

Four racers are running to the right. Yanis is twice as fast as Bora, Romina is 1.5 times as fast as Yanis and Globi is twice as fast as Yanis. They start in the positions pictured below. Which of the following statements are true?



- A Globi reaches the finish line before Romina.
- B Yanis crosses the finish line first.
- C Bora is the last one to reach the finish line.
- D Two racers cross the finish line at the same time.

Answer: B, C

Problem 20.

For two numbers $a > b$ from the list 3, 9, 15, 21, 27, what can you say for certain?

- A $a + b$ is even.
- B $a + b$ is divisible by 3.
- C $a - b$ is divisible by 6.
- D $a + b$ is *not* divisible by 7.

Answer: A, B, C

Problem 21.

Four real numbers a, b, c and d satisfy $a < b < c < d$ and $\frac{1}{b} < \frac{1}{a} < \frac{1}{d} < \frac{1}{c}$. Which of the following expressions must be positive?

- A $a + b + c + d$
- B $a - b - c + d$
- C $-a - b - c + d$
- D $a - b + c - d$
- E $-a + b + c - d$

Answer: C

Problem 22.

Jana goes to the MO-Mensa to have lunch. There are some meat dishes and some vegetable dishes (at least one of each type). A menu consists of either a meat dish, a vegetable dish, or both a meat dish and a vegetable dish. Jana computes that there are n possible menus. What could n be?

- A 2
- B 4
- C 6
- D 8
- E 10

Answer: D

Problem 23.

What is the largest prime divisor of $(12^{12} - 12^{11} + 12^{10})$?

- A 3
- B 7
- C 11
- D 19
- E 23

Answer: D

Problem 24.

A 2×3 rectangular paper is divided into six squares, numbered as shown below. Noelia folds the paper a total of three times, such that the six squares form a stack with square number 1 on top. Which square cannot be on the bottom?

- A 2
- B 3
- C 4
- D 5
- E 6

1	2	3
4	5	6

Answer: D

Problem 25.

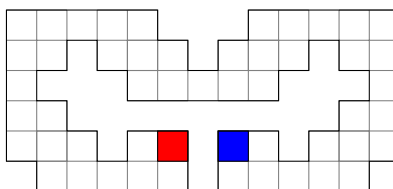
Find the smallest integer n such that exactly 2 of the following 5 statements are false:

1. n is divisible by 7.
2. n is divisible by 12.
3. n is divisible by 14.
4. n is divisible by 21.
5. n is divisible by 24.

Answer: 42

Problem 26.

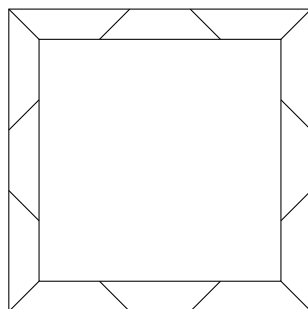
Rösti is walking through a cave. Starting at the red square and not entering the same square more than once, in how many ways can Rösti travel to the blue square?



Answer: $2^{12} = 4096$

Problem 27.

The frame of a painting has the form of a 105 cm by 105 cm square with a 95 cm by 95 cm square removed from its centre. The frame is built out of isosceles trapezoids with angles measuring 45° and 135° . Each trapezoid has one base on the frame's outer edge and one base on the frame's inner edge. Each outer edge of the frame contains an odd number of trapezoid bases that alternate long, short, long, short, etc. What is the maximum possible number of trapezoids in the frame?



Answer: 76

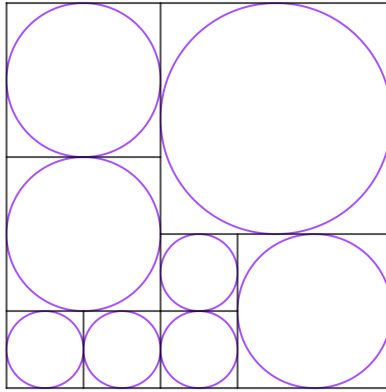
Problem 28.

What is the smallest integer $n > 3$ such that the sum and the product of the first n positive integers end in the same digit?

Answer: 15

Problem 29.

Arnaud divides his square field into smaller square plots to pass on to his grandchildren. Each one of them draws a circle as large as possible in their plot, as shown in the picture.



Which of the following statements are true, no matter how Arnaud divides his field?

- A The total area contained inside the circles is the same.
- B The total perimeter of the circles is the same.
- C The sum of the radii of the circles is the same.
- D The diagonals of the big field contain one or more centres of the circles.

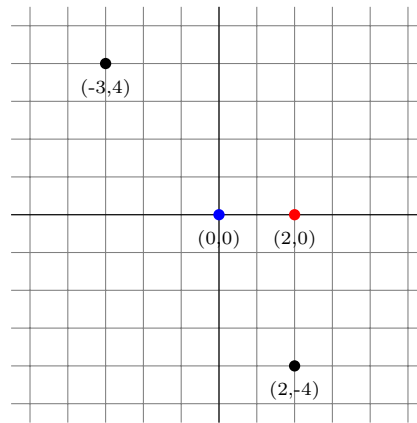
Answer: A, D

Problem 30.

Ricardo is playing with his weird compass on the coordinate plane. It has a blue leg, situated on the origin, and a red leg on $(2, 0)$. The distance between the two legs cannot be changed, and Ricardo moves his compass in the following manner: he fixes one of the legs, and moves the other leg to another lattice point.

Which of the following points can he reach with the red leg?

- A $(-3, 4)$
- B $(12, 22)$
- C $(20, 24)$
- D $(2, -4)$



Answer: B, D

Problem 31.

In MO-Canton, there are six towns such that no three towns are on a line. Annalena wants to connect as many pairs of towns as possible by straight roads such that no two roads cross. It turns out that, after having drawn M roads, she cannot make any more moves. Which of the following values cannot be equal to M , no matter how the towns are placed?

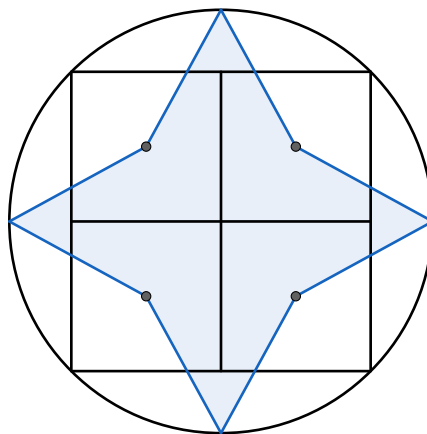
- A 8
- B 9
- C 10
- D 11
- E 12

Answer: A

Problem 32.

Given that the radius of the circle in the image below is 1, what is the area of the blue gem?

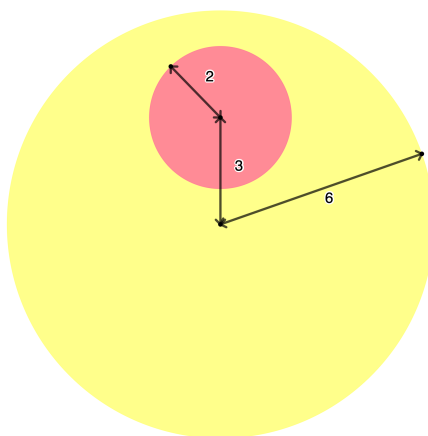
- A $\frac{3}{2}$
- B $\sqrt{2}$
- C $\frac{\sqrt{2}}{2} + \frac{3}{4}$
- D $\frac{5}{4}$
- E $\frac{1+\sqrt{2}}{2}$



Answer: B

Problem 33.

For Aisha's birthday, Quirin prepared a cake in the form of a cylinder whose top has a radius of 6. He added some icing, in the form of a disk with a radius of 2. The centre of the icing and the centre of the top are 3 apart.



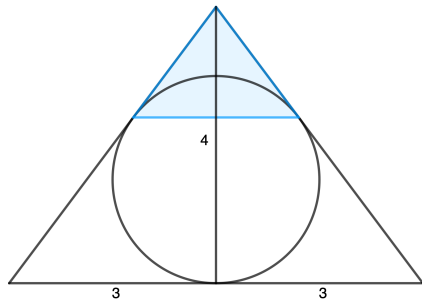
Raphi is very greedy: he wants to cut the cake into two parts using only one vertical stroke of a knife, and then steal one of the pieces. He also wants to make sure that both of the pieces have the same amount of icing. How large can the area on the top of his piece be, at most?

- A $36\pi - 6\sqrt{3}$
- B 24π
- C $24\pi + 9\sqrt{3}$
- D 18π
- E $24\pi + 6\sqrt{3}$

Answer: C

Problem 34.

A circle is inscribed in the following triangle, with height 4 and base 6.



What is the area of the blue triangle?

- A 12
- B $\frac{24}{5}$
- C $\frac{48}{5}$
- D $\frac{48}{25}$
- E $\frac{96}{5}$

Answer: D

Problem 35.

Julia has written down 6 distinct positive integers. She notices that

- the sum of any 3 of them is divisible by 3,
- the sum of any 4 of them is divisible by 4,
- the sum of any 5 of them is divisible by 5.

What is the smallest possible sum of all 6 numbers?

Answer: 906

Problem 36.

Marco the kangaroo starts on the red block and wants to jump to the blue block without falling into the gaps. In every jump, he can jump 1 or 2 blocks. In how many different ways can Marco get from red to blue?

A possible example of Marco's first 4 jumps is shown in the picture.



Answer: 2025

Problem 37.

David has 2025 coins in a row. If he removes the leftmost third of those coins, at least half of the remaining coins show heads. The same is true if he removes the rightmost 40%. At least how many coins show heads?

Answer: 743

Problem 38.

Paul draws a convex n -gon. All internal angles have integer degree values less than 180° , and they are all different. What is the largest possible value of n ?

Answer: 26

Problem 39.

On an unknown planet are 13 angels. For each pair of angels A and B , either A supervises B or B supervises A . One angel is named Viera.

Determine if the following statements are necessarily true, or not.

- A There is one angel which supervises every other.
- B Viera is supervised by at least 4 other angels.
- C There is one angel which supervises at least 6 other angels.
- D At least 12 angels are being supervised.

Answer: C, D

Problem 40.

Louis and Mark are playing a game. They start with two distinct positive numbers on the blackboard, and each step consists of the following operations:

- First, Louis doubles one of the integers on the blackboard.
- Then, Mark erases a number x and replaces it with two positive integers summing to x .

Which of the following statements are true?

- A Louis can make a number on the blackboard as large as he wants.
- B Mark can always guarantee that 2025 will appear on the blackboard at some point in time.
- C Louis can ensure that there are never more than one hundred 1s on the blackboard.
- D By cooperating, Mark and Louis can make all the numbers on the blackboard equal at some point in time.

Answer: D